ModelShield

A conversational AI service with separate, live extraction monitoring.

CHAT. OBSERVE.

RESTRICT THE BURST.

Ask real questions on the chatbot page. Watch the requests and red flags on the operator dashboard. The API can withhold answers from a suspicious key while other clients continue.

ModelShield Team

TWO PAGES, ONE PROTECTED SERVICE

Chat: http://127.0.0.1:8765/chat
Jury dashboard: http://127.0.0.1:8765/dashboard#requirements
Run: python enthi.py
First-time chat setup: python -m scripts.setup_chat

Local Qwen2.5-0.5B-Instruct, FastAPI, llama.cpp, three request-side signals, per-key restrictions, and retained incident evidence. The classifier lab remains available for the earlier copying experiments.

Watch how access is used

A valid API key can be used to collect answers for training an imitation.

The problem

Model extraction collects a model's outputs through repeated queries. An attacker does not necessarily need to compromise the server. This prototype looks for suspicious collection patterns while requests are arriving.

Who could deploy it

An AI service operator could place these checks between authenticated customers and a model runtime. It adds request-level explanations and selective answer withholding to API access controls. Production use would require customer-specific validation and durable shared state.

The customer experience

A separate chatbot provides short conversations and batches of up to 30 questions. It receives no operator risk scores. The dashboard shows requests, replies, restrictions, and red flags. Its Requirements & evidence view maps all five deliverables to working features and scoped measurements.

The demonstration

Send a few ordinary messages, then send a rapid batch of closely varied questions from a test client. Show the alert, its contributing signals, and HTTP 429 responses. A second client should still receive answers. Starting a new conversation does not clear the key's penalty.

The claim we can support

The prototype detects and restricts the demonstrated querying patterns. It does not prove malicious intent or guarantee prevention of cloning. Legitimate bulk traffic can look identical, and slow extraction can evade the current rules.

Check before answering

Chat signals use arrival timing and input wording; no sentiment or invented LLM confidence.

KEY > VALIDATE > SCORE > GATE > LOCAL MODEL > RECHECK > ANSWER

Rate / up to 45 points

Counts arrivals in the preceding 10 seconds. The contribution rises from zero at four requests to 45 at sixteen. Requests are counted before model generation, including requests rejected by the behavioral gate.

Diversity / up to 20 points

Mean pairwise cosine distance across the latest 32 hashed character 3-5-gram vectors. This measures lexical variation, not semantic meaning.

Repetition / up to 35 points

The larger of duplicate share and the share of consecutive pairs with cosine similarity at least 0.80. Diversity and repetition are scaled by sample count and recent activity, each reaching full weight at eight.

Speed bumps and restrictions

Below 35: normal. 35-39.9: watch. Risk 40-49.9 spaces generation starts by one second; 50-59.9 by three seconds. At 60+: throttle for five seconds; at 80+: block for 30 seconds. Server checks run before inference and before releasing an answer. Live waiting and actual delay appear on the dashboard.

Operator-approved bulk passes

The dashboard grants up to 30 requests for 1-60 minutes, with a concurrency cap. Approved volume receives rate credit; strong repetition removes it. SQLite preserves usage and revocation across restarts. The operator can run three live scenarios at /dashboard#protection. Fixed integration checks do not prove improved copying resistance.

Runtime and audit

The CPU model runs in an owned, authenticated loopback llama.cpp process with its web UI disabled. There are two generation slots and a bounded queue. The operator-only SQLite chat audit retains 5,000 questions and replies; live alerts preserve the signals at the decision.

Scope: one server process. Runtime windows and penalties reset on restart; audit records survive. The tiny model can make mistakes. These chat heuristics are separate from the classifier policy and have no cross-key correlation component.

Measured behavior and limits

Real-model browser checks and synthetic request-pattern checks answer different questions.

ATTACKS FLAGGED

66.7%

24 / 36 development sessions

FALSE POSITIVES

50%

12 / 24 benign sessions

VISIBLE ALERT

7.01s

One real browser burst run

Normal conversation	0 / 12 flagged
Legitimate bulk questions	12 / 12 flagged
Repeated extraction probes	12 / 12 flagged
Diverse extraction queries	12 / 12 flagged
Slow extraction probes	0 / 12 flagged

Method: five profiles across seeds 700-711, 30 arrivals per session, fixed policy, virtual time. Detection = flagged attack sessions / attack sessions. False positives = flagged benign sessions / benign sessions. These are development scenarios, not independent holdout estimates. Truth labels never enter the detector.

Failure cases: legitimate bulk questions use the same observable timing and inputs as diverse extraction, so all are flagged. All slow extraction sessions are missed. A faster alert does not establish that a clone became harder to train.

Real integration: 12 browser checks passed with the installed language model: real replies and follow-up context, separate operator authentication, 30-request enforcement, another key still answering, immutable incident evidence, export, mobile layout, and no browser JavaScript errors. The alert time includes browser polling and is one observed run.

Reproduce: python -m scripts.chat_check; python -m scripts.evaluate_chat; python -m pytest
Evidence: reports/chat-check.json and reports/chat-evaluation.json. Earlier classifier measurements appear separately on page 5.

Classifier copying results

Earlier sentiment experiments; these results do not measure chatbot protection.

COPYING THROUGH THE API

Same seed and candidate order; fresh detector state. Up to 120 attempts and 8 seconds per run. A second classifier learns released labels only. The evaluator compares labels on 90 separate sentences. Agreement is imitation, not a percentage stolen.

MODE	ANSWERS / TRIES	AGREEMENT	CUSTOMERS
unprotected	120 / 120	75.6%	16 / 16
protected	30 / 120	80.0%	16 / 16
protected retry after	28 / 32	80.0%	16 / 16

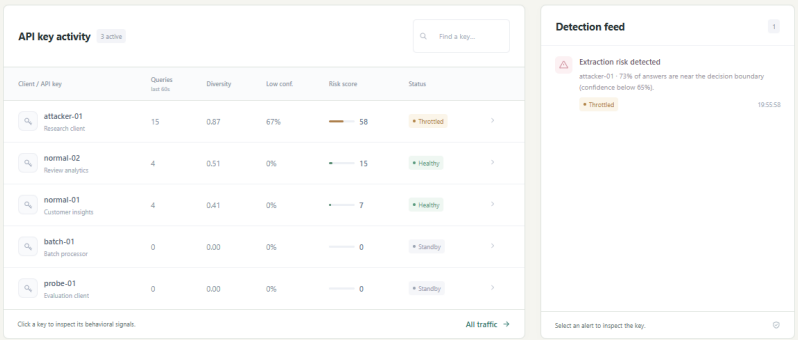
Interpretation: Protection did not lower copy agreement in this measured run. Fewer answers alone do not establish a weaker copy. A patient collector can wait for more answers. These short, synthetic experiments do not establish protection for other models or real customer traffic.

CALIBRATED DEFENSE COMPARISON

All enabled configurations meet the same 5% calibration false-positive target. Separate settings are frozen before challenge evaluation. The rate limiter has its own request capacity; both behavioral variants use equal two-second throttles. Production settings stay unchanged.

disabled	0/30 attacks detected	0/18 benign flagged
rate only	12/30 attacks detected	0/18 benign flagged
four signals	12/30 attacks detected	0/18 benign flagged
full	12/30 attacks detected	0/18 benign flagged

The full detector did not exceed the rate limiter's detection rate on this challenge set. Scenarios were authored here, not independently by a teammate. Settings were not retuned to improve these results. Inspect the per-scenario report for misses and false positives.



Live alert evidence is stored with its timestamp, signals, decision and policy.